

User Manual

Field Vision Analytics

A Data-Driven Approach to America's Game

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Introduction

Welcome to *Field Vision Analytics*, a comprehensive platform designed to help football teams, coaches, and analysts make sense of game performance data. Whether you are preparing for a game, analyzing team trends, or exploring player performance, this application provides a clear, visual, and interactive way to understand complex and messy data.

Key Features

- **Data Uploads:** Easily import your game and player data using simplified CSV templates
- **Interactive Dashboards:** Explore statistics, team leaders, and performance trends through intuitive charts and tables
- **Custom Analytics:** Filter and narrow down to specific players, position groups, or games to get the insights you need
- **Summarized Statistics:** Be able to see the season at a glance with the stat cards
- **Exports:** Download visualizations for sharing or presentation

Who Should Use FVA

- Coaches looking to track and evaluate team performance
- Analysts studying player and team trends
- Students or researchers exploring sport analytics
- Anyone interested in visualizing and interpreting football data

This manual will guide you through every part of the application, from uploading your first dataset to navigating the dashboards, downloading visualizations, and troubleshooting common issues. By the end of this manual, you should feel confident using the full capabilities of *Field Vision Analytics* to turn raw, complicated data into actionable insights.

Getting Started

This section walks you through opening the application and utilizing every aspect of the dashboard. It also lists the system requirements for running the application.

System Requirements

FVA is a web-based application and can be accessed through most modern web browsers. For the best experience, the following are recommended:

- **Browser:** Google Chrome, Microsoft Edge, or Apple Safari
- **Internet Connection:** Stable connection for uploading and processing data
- **Screen Size:** Desktop, laptop, or tablet display recommended for full dashboard visibility

Accessing the Application

There are two different ways to access FVA. The first way can only be done by Truman students and faculty on the Truman State University internet network. It is hosted on Truman's plasma server, which is only accessible when on Truman's internet. The steps for that method are:

1. Open your preferred web browser
2. Navigate to <http://plasma.truman.edu:3838/bk47466/FieldVisionAnalytics/>
3. Wait for the application to fully load before interacting with any features

The second way can be done by anybody with a standard internet connection. The steps for that method are as follows:

1. Open your preferred web browser
2. Navigate to <https://gracie-lovell.shinyapps.io/FieldVisionAnalytics/>
3. Wait for the application to fully load before interacting with any features

Navigation Overview

This application is organized into a simple and intuitive layout:

- **Upload Section:** This is where users import their data files (CSV templates). Each dataset has its own upload area

- **Content/Dashboard Section:** Once the data is uploaded, this section displays visualizations, charts, stat cards, and leaderboards based on your data
- **Help Button:** Available throughout the application, the help menu provides access to the CSV templates and this user manual

Navigation buttons are typically located at the top of the screen, allowing users to move between sections easily.

First Time Setup

To get started with your first analysis:

1. The application will open in the Upload Section
2. Navigate to the Help Menu located by the “New Here?” button in the header
3. Download the six CSV templates required for full analysis
4. Fill in the templates with your data using the required format
5. Download the completed CSV files to your computer
6. Upload each file to the corresponding upload area
7. Once all required data is uploaded, navigate to the Content Section

Tips for First-Time Users

- Ensure all column names in your CSV files match the templates exactly
- Avoid leaving any fields blank
- Upload all datasets if you want to utilize the dashboard’s full computing power
- If something does not load, double-check your data formatting or refresh the application

Uploading Data

Overview

The Upload Section allows users to import their own team and player data into FVA. The application uses this data to generate all visualizations, leaderboards, tables, and metrics. To ensure the user uses the application to its fullest potential, it is recommended that all files be

uploaded into the program before continuing to the Dashboard. However, it is possible to only upload specific files if the analysis the user is using only requires a subset of the data.

Supported File Types

FVA only accepts CSV files at this time. This means that all files need to have the extension “.csv”. While Excel (.xlsx) may be used to create and edit the data, it must be saved as CSV files before being uploaded to the application.

Required Data Files

Users are expected to upload multiple datasets, each corresponding to a different area of analysis:

- **Passing Data:** Passing yards, touchdowns, and interceptions
- **Rushing & Receiving Data:** Rushing & receiving yards, touchdowns, and fumbles
- **Defensive Data:** Tackles, sacks, interceptions, and forced fumbles
- **Special Teams Data:** Field goal percentage, returns, and yardage
- **Roster Data:** Player-by-player statistics
- **Schedule Data:** Game results and scores

Data Formatting Requirements

To ensure successful uploads:

- Column names must match the template exactly
- Do not remove or rename columns
- Each row should represent one observation
- Avoid empty rows or columns
- Do not include merged cells or formulas

Best Practices

- Start by downloading and using the provided CSV templates
- Enter data carefully and consistently
- Test uploads with a small dataset first
- Save a backup of your original data before uploading

After Uploading

Once all required files are successfully uploaded:

1. Press one of the “Launch Dashboard” buttons to navigate to the Content Section
2. Explore charts, leaderboards, and analytics generated from your data
3. Use filters and controls to customize your view

Data Dictionary

Below is a description of each template, its purpose, and its variables:

Passing

Contains quarterback and passing statistics; Used to identify top passers and analyze passing performance.

Column Name	Variable Definition	Requested Format
Player	Name of the player	Character String
Age	Age of the player	Numeric
Pos	Position of the player	Character String
Games Played	The number of games that the player has played	Numeric
Games Started	The number of games that the player has started in	Numeric
Completions	The number of completed passes the player has thrown	Numeric
Attempts	The number of attempted passes the player has thrown	Numeric
CompPerc	How many completed passes were made out of the number of attempted passes (Completions / Attempts)	Numeric
Yards	The number of yards that the player has thrown for	Numeric

Touchdowns	The number of passes that have resulted in a touchdown	Numeric
TouchdownPerc	How many touchdowns were completed out of the number of attempted passes (Touchdowns / Attempts)	Numeric
Interceptions	The number of times a player of the opposing team has caught the ball thrown by the player	Numeric
IntPerc	How many interceptions were made out of the number of attempted passes (Interceptions / Attempts)	Numeric
FirstDowns	The number of times a pass has resulted in a first down	Numeric
SuccessRate	A successful pass gains at least 40% of yards required on 1st down, 60% of yards required on 2nd down, and 100% on 3rd or 4th down. Denominator is pass attempts + times sacked	Numeric
Sacks	The number of times the opposing team has sacked a passer	Numeric

Rushing & Receiving

Contains offensive skill position statistics; Used to determine rushing and receiving leaders and evaluate offensive production.

Column Name	Variable Definition	Requested Format
Player	Name of the player	Character String
Age	Age of the player	Numeric
Pos	Position of the player	Character String

Games Played	The number of games that the player has played	Numeric
Games Started	The number of games that the player has started in	Numeric
Rush Attempts	The number of times the player attempted to run with the ball	Numeric
Rushing Yards	The number of yards that the player was able to get by running the ball	Numeric
Rushing Touchdowns	The number of touchdowns that the player was able to get by running the ball	Numeric
Rushing Firstdowns	The number of first downs that the player was able to get by running the ball	Numeric
Rushing SuccessRate	A successful rush gains least 40% of yards required on 1st down, 60% of yards required on 2nd down, and 100% on 3rd or 4th down. Denominator is rushing attempts.	Numeric
Targets	The number of times the player attempted to catch the ball	Numeric
Receptions	The number of times the player was able to catch the ball successfully	Numeric
Reception Yards	The number of yards that the player was able to get by catching the ball	Numeric
Receiving Touchdowns	The number of touchdowns that the player was able to get by catching the ball	Numeric

Receiving Firstdowns	The number of first downs that the player was able to get by catching the ball	Numeric
Receiving SuccessRate	A successful reception gains at least 40% of yards required on 1st down, 60% of yards required on 2nd down, and 100% on 3rd or 4th down. Denominator is targets.	Numeric
Fumbles	The number of times a fumble was lost and recovered by own team	Numeric

Defense

Contains defensive player statistics; Used to identify defensive leaders.

Column Name	Variable Definition	Requested Format
Player	Name of the player	Character String
Age	Age of the player	Numeric
Pos	Position of the player	Character String
Games Played	The number of games that the player has played	Numeric
Games Started	The number of games that the player has started in	Numeric
Interceptions	The number of passes intercepted by the player	Numeric
IntYards	The number of yards that resulted from interceptions caught	Numeric
Interception Touchdowns	The number of interceptions that resulted in a touchdown	Numeric
Passes Defended	The number of passes blocked by the player	Numeric

Forced Fumbles	The number of times the player forced a fumble to occur, can be recovered by either team	Numeric
Fumbles	The number of times a fumble was lost and recovered by own team	Numeric
Fumbles Recovered	The number of times the player recovered a fumble made by either team	Numeric
Fumble Yards	The number of yards that recovered fumbles were returned	Numeric
Fumbles Resulted TD	The number of times a recovered fumble resulted in a touchdown	Numeric
Sacks	The number of times a player tackled the quarterback behind the line of scrimmage while they still had possession of the ball	Numeric
Combined Tackles	Solo + Assisted Tackles	Numeric
Solo Tackles	The number of tackles made by the player independently	Numeric
Assisted Tackles	The number of tackles made by the player with another player's assistance	Numeric
Tackles for Loss	The number of times the player tackled the opposing team's player behind the line of scrimmage, resulting in a negative yardage	Numeric
QB Hits	The number of times the player tackled the quarterback	Numeric
Safeties	The number of times the player tackled the offensive player inside their endzone	Numeric

Special Teams

Contains kicking, punting, and returning statistics; Used to identify leaders in special team categories

Column Name	Variable Definition	Requested Format
Player	Name of the player	Character String
Age	Age of the player	Numeric
Pos	Position of the player	Character String
Games Played	The number of games that the player has played	Numeric
Games Started	The number of games that the player has started in	Numeric
Punt Returns	The number of times the player returned a punt	Numeric
Punt Return Yardage	The number of yards that the player got from a punt return	Numeric
Punt Return for TD	The number of times a punt return resulted in a touchdown	Numeric
Kickoff Returns	The number of times the player returned a kickoff	Numeric
Kickoff Return Yardage	The number of yards that the player got from a kickoff return	Numeric
Kickoff Return for TD	The number of times a kickoff return resulted in a touchdown	Numeric
Field Goals Att	The number of times the player attempted to kick a field goal	Numeric
Field Goals Made	The number of times the player successfully made a field goal	Numeric

Field Goal Perc	How many field goals were completed over the number of field goals attempted (Field Goals Made / Field Goals Att)	Numeric
Extra Points Att	The number of times a player attempted to kick an extra point	Numeric
Extra Points Made	The number of times a player completed an extra point kick	Numeric
Extra Point Perc	How many extra points were completed over the number of extra points attempted (Extra Points Made / Extra Points Att)	Numeric
Kickoffs	The number of times a player has kicked the ball in a kickoff	Numeric
Kickoff Yards	The number of yards that the player received in their kickoff kicks	Numeric
Touchbacks	The number of times a kick has landed and been downed by the opposing team in the endzone	Numeric
Touchback Perc	The percentage of times that a kickoff was a touchback	Numeric
Punts	The number of times a player has kicked the ball for a punt	Numeric
Punt Yardage	The number of yards that the player received in their punts	Numeric

Roster

Contains basic player statistics; Used to gather knowledge about players and their careers

Column Name	Variable Definition	Requested Format
Number	Number of the player	Numeric, could be “NoNum”
Player	Name of the player	Character String
Age	Age of the player	Numeric
Pos	Position of the player	Character String
Games Played	The number of games that the player has played	Numeric
Games Started	The number of games that the player has started in	Numeric
Weight	Weight of the player	Numeric (in lbs)
Height	Height of the player	Numeric (in ft-in)
Years	The number of years the player has been in the league	Numeric, a rookie, or someone with 0 years is written as “Rook”

Schedule

Contains the game schedule of the team; Used to generate score margin charts, game trends, and win/loss analysis

Column Name	Variable Definition	Requested Format
Game Number	The game number for the team (does not skip with bye)	Numeric
Week	The week of the season (does skip the bye week)	Numeric
Date	The date of the game	Date (YYYY-MM-DD)
Day	The day of the week on which the game was played	Character String

Location	If the game was played at home or away	Character String (Home/Away)
Opponent	Who was the opponent of the game	Character String
Result	The result of the game	Character String (Win, Loss, or Tie)
Points For	The number of points the team scored	Numeric
Points Against	The number of points the opposing team scored	Numeric
Overtime	Whether the game went into overtime or not	Character String (Yes/No)

Dashboard & Analytics

Overview

The Dashboard is the core of Field Vision Analytics. After uploading data, this area transforms raw statistics into clear, interactive visualizations and summaries. Users can explore team performance, compare statistics, and identify key trends across offense, defense, and special teams.

Navigation Between Units

The dashboard is organized into six primary tabs:

- Overview
- Offense
- Defense
- Special Teams
- Roster
- Schedule

Users can switch between these tabs using the navigation buttons. Each tab displays its own set of relevant statistics and tables.

Key Visualizations

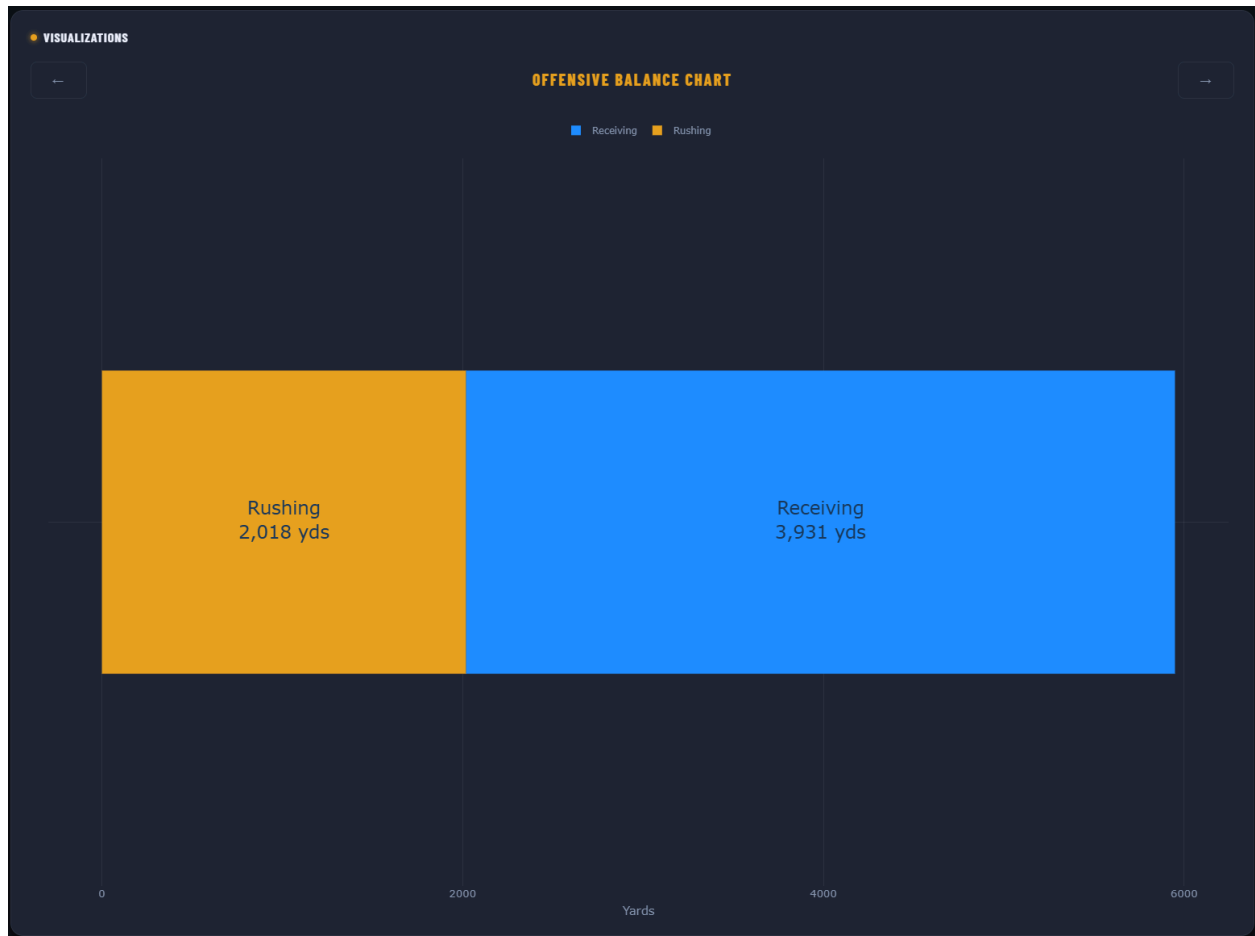
The first visualization is a simple scatterplot that compares offensive output vs. defensive performance. Each point represents a single game, and they are labeled with the opponent for easy distinction.



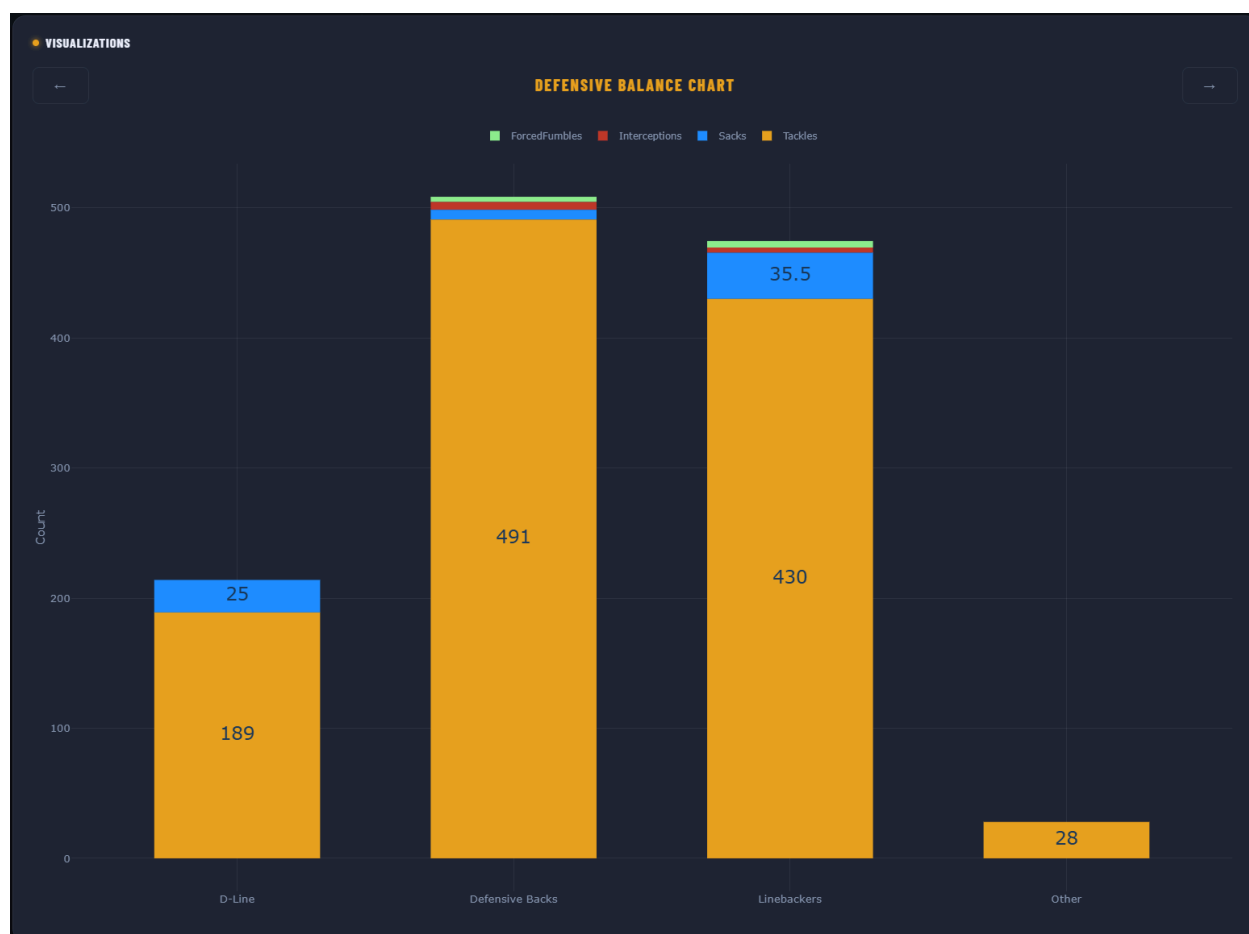
The next visualization is a score margin chart that displays the point differential for each game. The positive values indicate wins and the negative values indicate losses. The bars are color-coded to distinguish between results easily.



The third visualization is called an offensive balance chart. Its main goal is to visualize the ratio between rushing and receiving yards. This helps determine how balanced or one-dimensional the offense is.



Then, we have the same type of chart but for the defense. The defensive balance chart compares different types of defensive production (tackles, sacks, etc.) against position groups. This visualization shows how defensive impact is distributed across position groups.



The final visualizations are the player leaderboards. These tables display top-performing players in selected units of each aspect of the football team. This gives the players a spotlight for leading their team in a certain statistic.

LEADERS

OFFENSE

PASSING YARDS	Bo Nix	3,931
RUSHING YARDS	J.K. Dobbins	772
RECEIVING YARDS	Courtland Sutton	1,017

LEADERS

DEFENSE

TACKLES	Alex Singleton	135
SACKS	Nik Bonitto	14
INTERCEPTIONS	Ja'Quan McMillian	2

LEADERS		SPECIAL TEAMS	
KICKER	Wil Lutz		28
PUNTER	Jeremy Crawshaw		3,573
KICK RETURNER	Marvin Mims		658
PUNT RETURNER	Marvin Mims		452

Interactivity

The dashboard includes several interactive features:

- **Hovering:** You can view additional details on the charts if you hover over them
- **Dynamic Updates:** Visualizations automatically update when new data is uploaded
- **Filtering:** Many different types of filters can be used to narrow down the data for better viewing or querying
- **Searching:** You can search any of the data tables in the database to find specific players or observations
- **Responsive Layout:** The layout adjusts to different screen sizes

Limitations

The visualizations are only as accurate as the uploaded data. Any missing or incorrect data may impact results. Also, at this moment, this application does not perform any advanced predictive modeling techniques.

Technical Support

For help with any technical issues, please contact the developer at bk47466@truman.edu.

This user manual is maintained by the FVA development team.

Last Updated: April 2026